

Unit of Observation

Each row of the Spotify reviews dataset represents a single user-submitted review of the Spotify mobile application. A review consists of the user's written text feedback, a numeric star rating, and associated metadata such as the review date or app version if present. This dataset is used to analyze sentiment expressed in text reviews and compare it to the corresponding star rating.

Variable Descriptions

Below is a complete list of all variables analyzed in this project. For each variable, we provide the description, data type, possible values or range, and summary statistics where applicable.

review_text

Description:

The written text of the user's review of the Spotify app. This field contains the natural language feedback that is used for sentiment analysis.

Data Type:

Text (string)

Example Values:

- "I love the new update — it's so smooth!"
- "App crashes every time I open it"
- "Pretty good but ads are annoying"

Summary Statistics:

- Number of non-missing reviews: *[N non-missing]*
- Empty or blank reviews removed in preprocessing: *[N blank]*

Notes:

This variable is the primary input to sentiment analysis. Missing or empty entries were dropped prior to analysis.

rating

Description:

The star rating provided by the user for the Spotify app. Ratings serve as the numeric measure of user satisfaction.

Data Type:

Integer (1 to 5)

Possible Values:

1, 2, 3, 4, 5

Summary Statistics:

Metric	Value
Count	<i>[N total reviews]</i>
Mean	<i>[μ]</i>
Median	<i>[median]</i>
Mode	<i>[mode]</i>
Ratings Range	1–5
Rating missing or invalid	<i>[N invalid]</i>

Notes:

Only ratings between 1 and 5 were retained during data cleaning. Ratings outside this range or missing were omitted from analysis.

review_length_words (engineered)**Description:**

The word count of the review text, created during preprocessing to understand review length distribution and its relationship to sentiment.

Data Type:

Integer (≥ 0)

Summary Statistics:

Metric	Value
Min	<i>[min]</i>
Max	<i>[max]</i>
Mean	<i>[mean]</i>
Standard Deviation	<i>[std]</i>

Notes:

This variable helps capture how review length correlates with sentiment scores.

review_length_chars (engineered)

Description:

The total number of characters in a review, used as an alternative measure of text length.

Data Type:

Integer (≥ 0)

Summary Statistics:

Metric	Value
Min	<i>[min]</i>
Max	<i>[max]</i>
Mean	<i>[mean]</i>
Standard Deviation	<i>[std]</i>

vader_neg, vader_neu, vader_pos

Description:

VADER sentiment analysis sub-scores indicating the proportion of negative, neutral, and positive sentiment within the text review.

Data Type:

Float (0.0–1.0)

Summary Statistics:

Component	Min	P25	Median	P75	Max
vader_neg	<i>[min]</i>	<i>[p25]</i>	<i>[median]</i>	<i>[p75]</i>	<i>[max]</i>
vader_neu	<i>[min]</i>	<i>[p25]</i>	<i>[median]</i>	<i>[p75]</i>	<i>[max]</i>
vader_pos	<i>[min]</i>	<i>[p25]</i>	<i>[median]</i>	<i>[p75]</i>	<i>[max]</i>

Notes:

VADER (Valence Aware Dictionary and Sentiment Reasoner) is a lexicon-based sentiment tool optimized for social text.

vader_compound

Description:

The compound sentiment score from VADER, representing a normalized sentiment value between -1 (most negative) and $+1$ (most positive).

Data Type:

Float (-1.0 to $+1.0$)

Summary Statistics:

Metric	Value
Min	<i>[min]</i>
P25	<i>[p25]</i>
Median	<i>[median]</i>
Mean	<i>[mean]</i>
P75	<i>[p75]</i>
Max	<i>[max]</i>
Neutral Rate (-0.05 to 0.05)	<i>_[fraction]*</i>

Notes:

Compound scores within -0.05 to 0.05 are typically considered neutral according to VADER documentation.

Exploratory Figures and Tables

The following figures were created during data establishment and should be included in this appendix as visual supplements.

Figure A: Sentiment Distribution Across Review Length Categories

Description:

This figure shows the distribution of VADER compound sentiment scores across five review length categories for all 61,594 reviews.

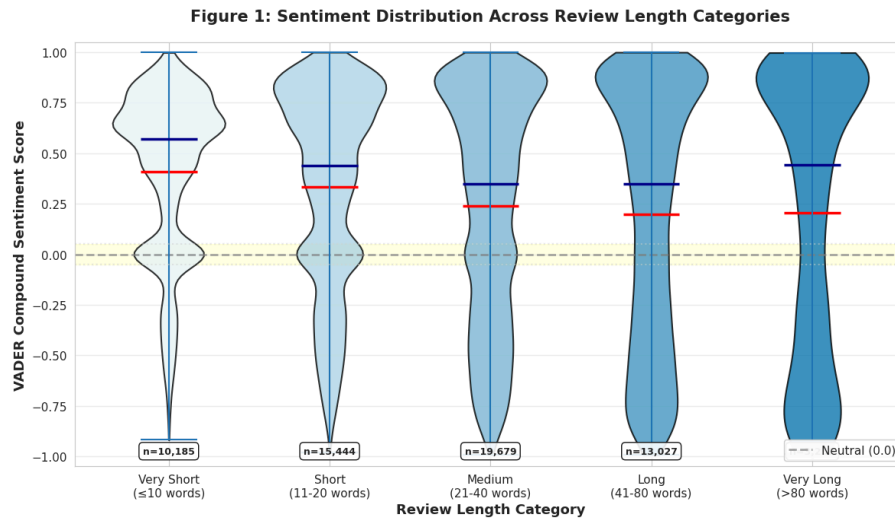


Figure B: Distribution of VADER Compound Sentiment Scores

Description:

This histogram displays the overall distribution of VADER compound sentiment scores across 61,594 reviews.

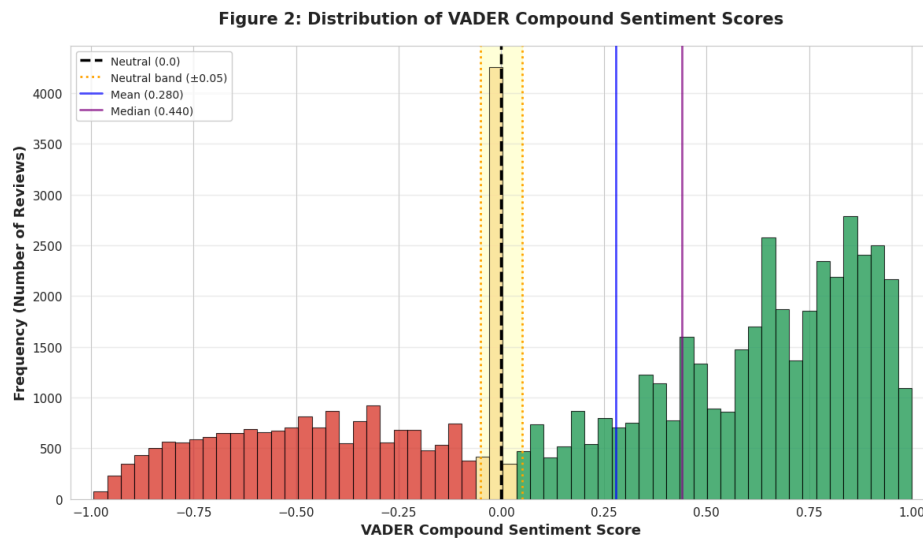
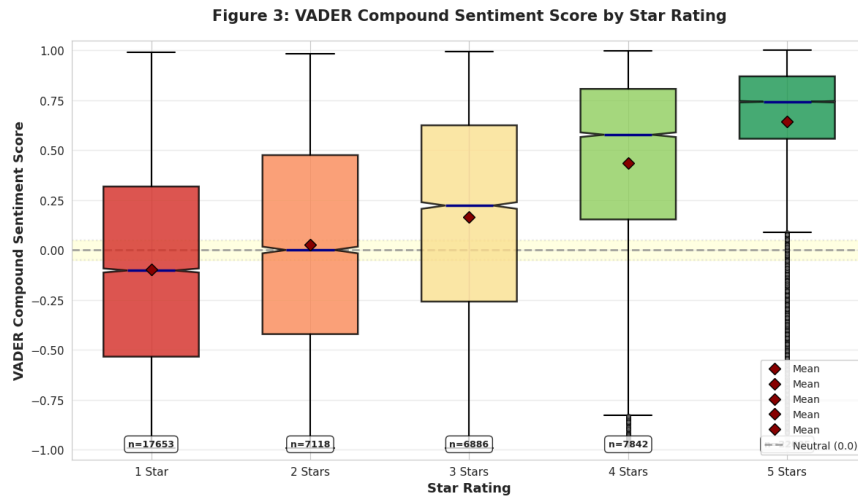


Figure C: Compound Sentiment by Star Rating

Description:

This boxplot shows compound sentiment scores grouped by star rating (1–5 stars).



Notes on Data Processing

- Reviews with missing text or missing star ratings were excluded from analysis.
- Ratings outside 1–5 were dropped.
- All variables were generated using scripts in the SCRIPTS folder (project1.ipynb and official_code.ipynb).
- Engineered variables (review_length_words, review_length_chars, and sentiment scores) were computed in preprocessing and sentiment scoring steps.